AP PHYSICS 1

Unit 1 • Worksheet 1.4A

Worksheet 1.4A • Reference Frames and Relative Motion

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

A train moves at 25 m/s. A passenger walks toward the front of the train at 2 m/s relative to the train. What is the passenger's speed relative to the ground?

(A) 23 m/s

(B) 27 m/s

(C) 25 m/s

(D) 2 m/s

Question 2

The same passenger now walks toward the back of the train at 2 m/s relative to the train. Her velocity relative to the ground is

(A) 23 m/s forward

(B) 27 m/s backward

(C) 23 m/s backward

(D) 2 m/s backward

Question 3

Car A moves north at 30 m/s and car B moves north at 20 m/s. What is the velocity of B relative to A?

(A) 10 m/s north

(B) 50 m/s north

(C) 10 m/s south

(D) 50 m/s south

Question 4

Two cars drive toward each other, each at 20 m/s relative to the road. What is the speed of one car relative to the other?

(A) 0

(B) 20 m/s

(C) 10 m/s

(D) 40 m/s

Question 5

A boat points straight across a river 60 m wide at 5 m/s relative to the water. The current is 2 m/s. How long does it take to cross?

(A) 12 s
(B) 30 s
(C) 8.6 s
(D) 11.1 s

Question 6

How far downstream does the boat in the previous question land?

(A) 12 m
(B) 24 m
(C) 60 m
(D) 30 m

Question 7

A passenger in a train moving at constant velocity drops a ball. For the passenger, the path of the ball is

(A) straight down
(B) a curve bending backward
(C) a curve bending forward
(D) horizontal

Question 8

For an observer standing beside the track, the path of the same ball is

(A) straight down
(B) a curve that moves forward with the train
(C) a curve that moves backward
(D) a horizontal line

Question 9 (Free response)

An airplane heads north at 200 m/s relative to the air. A wind blows east at 50 m/s.

(a) Find the magnitude and direction of the plane's velocity relative to the ground.
(b) How far north does the plane travel in one hour?
(c) In which direction should the pilot point the plane to fly due north? Explain.